

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027

Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

TherAlign Health, Founding Software Engineer Intern | *Remote* Jul 2026 – Present

- Founding engineer on a provider-facing **SMART on FHIR** clinical decision support app (Next.js + Firebase/GCP), building **OAuth2/PKCE** launch and token flows and **HIPAA-aligned handling of PHI** in the EHR prescribing workflow.
- Extending a serverless **Firebase Cloud Functions** backend where **Gemini** is constrained to synthesis while the backend deterministically controls eligibility and evidence, keeping AI output verifiable and guardrailed rather than a black box.

Culture & Morality Lab, UMass Amherst, Lab Assistant | *Amherst, MA* Jul 2026 – Present

- Sole engineer building the lab's text-analysis platform (**Python**, **FastAPI**, React) end to end, owning architecture and all technical decisions.
- Built a YAML model registry and a structured, machine-readable data-quality warning system with deterministic checks that report uncertainty instead of guessing; grew the backend to **40 hermetic tests** running in CI.

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Owned backend security and reliability for the multi-tenant eTriage stack, implementing **AWS KMS** encryption for sensitive clinical data, hardening authentication and rate limiting, and remediating **20+ VAPT findings** with regression-tested fixes.
- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly submissions** via Mayden iaptus REST integrations; contributed to **\$340K+** in revenue.
- Diagnosed and resolved production issues across tenants, monitoring logs during releases and client integrations and shipping targeted fixes for tenant-specific incidents to keep the multi-tenant service stable.

Projects

Refusal Decay – LLM Safety / Adversarial Robustness (GitHub) PyTorch, Llama-3.1-8B, AdvBench

- Studied how prefilling **attacks** weaken safety refusal in Llama-3.1-8B-Instruct, extracting the residual-stream refusal direction via difference-in-means (behavioral refusal dropped **0.92 to 0.32** at prefill $k=3$) with causal activation patching, bootstrap 95% CIs, and McNemar's test.

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a Python/FastAPI agentic web-discovery service with deterministic extraction, LLM fallback, and **cell-level provenance** for every value; hardened with **150+ automated tests** and provider routing/fallback across LLM backends.

cf-ai-research-scout – Edge-Native AI Assistant (GitHub) TypeScript, Cloudflare Workers AI, Durable Objects, D1

- Built an edge-native AI research assistant on **Cloudflare Workers** with durable workflows (**WorkflowEntrypoint** + **Durable Objects**) that survive process restarts and tool failures, and **D1** for state, no external database.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages: Python, Go, TypeScript, JavaScript, C++, Bash, SQL

Security: VAPT remediation, authentication & authorization, AWS KMS encryption, rate limiting, audit logging, OAuth2/PKCE, secure-by-default

Backend & Web: Node.js, Express, FastAPI, React, Next.js, REST APIs, multi-tenant architecture

AI & LLM: LLM agents, RAG, LLM safety / adversarial robustness, cell-level provenance, PyTorch, sentence-transformers

Cloud & DevOps: AWS, GCP, Azure, Cloudflare Workers, Docker, Git, GitHub Actions

Open Source & Community

- Built **Underdogs Fitness** (Website), a production MERN platform with Stripe payments; live **3+ years**. - Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.